

POLITECNICO DI MILANO







Firm financial evaluations

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Distinguishing internal managerial decisions vs financial perspectives

Operational perspective (entrepreneur):

- Cover fixed costs
- Achieve profitability
- Scale revenues

Financial perspective (investor):

- What is my expected return?
- Is it sufficient for the risk?
- How much equity must I receive?

Break-even is necessary.
IRR is decisive.

Current situation

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Venture capital: a crucial backer of fast-growing businesses

More than a third of the €66.2bn worth of VC deals in Europe were in AI-related companies



Leading AI start-ups such as UK autonomous driving group Wayve have seen their valuations surge in deals over the past year © Charlie Bibby/FT

Ivan Levingston

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Some of those highly valued start-ups have emerged as hubs for producing new streams of entrepreneurs. Sweden's buy-now, pay-later pioneer Klarna is the leader with 66 companies linked to former employees, while Spotify is second with 61, according to a report from Accel.

"Now we have repeat entrepreneurs, repeat management teams," says Harry Nelis, a partner at Accel. "Europe has a really good mixture of potentially great venture opportunities especially when it comes to AI. There's a number of universities here in the UK, France and Germany that are leading places."

The rising class of AI start-ups are seeing their businesses expand at speeds that previously would have been unimaginable. That has made it more difficult for more established companies that appear slower growing in comparison.

"What we're seeing right now is every week we meet two to three companies that went from 0 to $10mn of revenues in the last year," says Carolina Brochado, a partner who leads venture and growth investing for EQT. "When that hyper growth becomes table stakes, it's very challenging."

Some of Europe's older start-ups, particularly in financial services such as the $75bn fintech Revolut, still remain attractive to investors after demonstrating they can build international businesses that continue to grow rapidly.

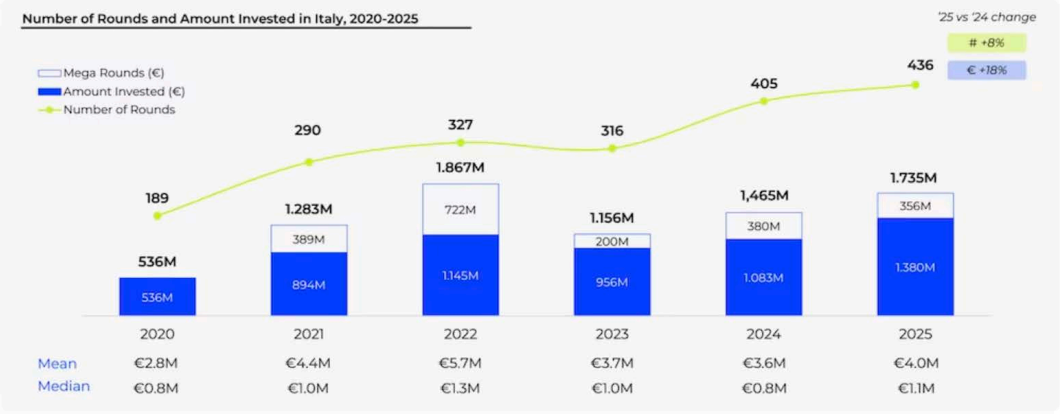
Current situation

Italian VC in 2025

- **€1,735M** raised
- **436** rounds
- **2** mega rounds amounting to **€356M**
- **53** Series A
- **14** Series B
- **5** Series C+
- **9** new VC funds raised a total of **€545M**
- largest round **€234M**[1] by **BENDING SPOONS**
- **31** Exits

3.1.1. VC in Italy: Historical Evolution by Year

2025 marks the highest year ever recorded in terms of number of rounds and surpasses 2024 in total capital invested, both including and excluding the impact of the two mega rounds closed in Q4-25



In brief: Carbon Equity launches co-investment fund

Co-Invest Fund I is aiming for a 25% IRR and will invest in climate tech companies, according to a LinkedIn post.

Charles Avery - 18 December 2025

Share A- A+ 100%



Jacqueline van den Ende, Carbon Equity CEO

Carbon Equity, the European manager that seeks to channel retail capital into climate investments, will invest directly into companies for the first time, as it launches a co-investment fund targeting climate tech companies.

Co-Invest Fund I is aiming for a 25 percent IRR, and will invest in at least 12 companies.

The investors is interested in the Internal Rate of Return (IRR)

The investor examines the cash flows (CF) of a firm and establish the expected value of the firm at the Exit, the moment in which the investor plans to sell the shares of its investment. When evaluations do not involve Exits, investors use a Continuing Value to determine the value of the expected stable levels of Cash for an explicit level of T years.

$$0 = -I_0 + \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t} + \frac{Exit_T}{(1+r)^T}$$

If the fund receives cash only at exit:

$$0 = -I_0 + \frac{Exit_T}{(1+r)^T}$$

Solving for r:

$$IRR = \left(\frac{Exit_T}{I_0}\right)^{\frac{1}{T}} - 1$$

The IRR is the return expected by the investor

Determining the value of a company

An investment Fund decides to evaluate an investment in a corporation. To get the shares of a corporation, the Fund promises its investors a return of around 30%. Accordingly, the required IRR to make that investment possible is 30%.

The Fund then evaluates whether and how the investment is convenient

A fund considers investing:

- Investment today: €3,000,000
- Holding period: 5 years
- Required IRR: 30%

Key question:

What must the company be worth in 5 years to justify this investment?

Determining the value of a company

Assuming no Cash Flow received within the 5 years of holding period, the Fund establishes the amount of money that should be earned out of that investment.

IRR constraint --> If the fund requires 30% annually:

$$Future\ Value = Investment \times (1.30)^5$$

$$(1.30)^5 = 3.713$$

$$Required\ exit\ value = 3M \times 3.713 = 11.14M$$

The fund must receive €11.14M at exit.

This is not a negotiation, this is an financial evaluation

Given the 11.4 M required at exit, what will be the future value of the corporation ?

Gordon Shapiro Model

Determine the value of the firm, based on the value of a share (price of a share).
The value (price) of a share is determined by a dividend discount model (DDM)

$$P_0 = \sum_{t-1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

P= price of a share
DIV_t = dividend expected from time t until infinite
r = rate of return required

Gordon Shapiro Model

- In practice, it is unrealistic to assume the availability of dividend flow estimates over an infinite time horizon, as required by the standard **DDM** (Dividend Discount Model).

- Typically, reliable short-term growth forecasts (3-5 years) are provided by financial service firms, such as the **Value Line Investment Survey** and the **Institutional Brokers' Estimate System (I/B/E/S),** which also regularly publish estimates for other key investor metrics. Consequently, the primary challenge for a financial analyst lies in estimating long-term dividend trends --> **GROWTH**.

- This issue is generally addressed by **assuming specific patterns in dividend behavior, known as dividend growth (*g*) paths, given that for any year *t*, dividend growth can be expressed as follows**:

$$g_t = \frac{DIV_t - DIV_{t-1}}{DIV_{t-1}}$$

Assuming perpetual yield over time

- We do assume to consider cash flows as they face similar patterns of perpetual growth over time

Gordon Shapiro Model

- The Gordon Growth Model, with one «g» level is suitable for 'mature' companies in known and established sectors, rather than those operating in 'emerging' sectors with profit rates exceeding the market average.

- In such cases, analysts suggest using a **multi-stage growth model.** The simplest version assumes distinct values for the growth rate, g.
  - The higher levels of «g» across periods of evaluation, the higher the precision of the evaluation

Gordon Shapiro Model

- The assumption is that the yield growth over time of firms' cash flows, for an infinite long term span, is fixed.



- g is constant over time and is lower than the cost of capital

Enterprise value

The growth component combines with the cost of capital for determining the value of a corporation

$$EV = \frac{FCFF_{t+1}}{WACC - g} = EV = \frac{FCF_{t+1}}{r - g}$$

$$FCF = k \cdot EBITDA$$

$$EV = \frac{k \cdot EBITDA_{t+1}}{r - g}$$

$$EBITDA_{t+1} = EBITDA_t \cdot (1 + g)$$

$$EV = \frac{k \cdot EBITDA_t \cdot (1 + g)}{r - g}$$

$$\frac{EV}{EBITDA_t} = \frac{k(1 + g)}{r - g}$$

- *EV:* Enterprise Value
- *k* = cash conversion ratio (% EBITDA → CF)
- *r* = required return (cost of capital / WACC)
- *g* = growth rate of EBITDA or CF, expression of the perpetual yield over time

Determining the value of the company – market multiple

- Market multipliers is a term applied to EBITDA, EBIT or to Cash Flows which help determining how many times a company worths its EBITDA / EBIT or Cash Flows.

- Multipliers assume growth projections, innovation trajectories, risk dynamics, sector criticalities or opportunities. It is mostly sector specific

- The multiplier can be given by the market standards for existing comparable transactions (mostly used), or can be calculated
  - To be calculated it requires to estimate:
    - g: % growth projection of cash flows
    - r: % discount rate of the cash flows
    - k: % of the EBITDA that can be considered cash (EBITDA * k)

Determining the value of the company – market multiple

$$Value = \frac{EBITDA \cdot k}{r - g}$$

$$\frac{Value}{EBITDA} = \frac{k}{r - g}$$

- If EBITDA $_{(t+1)}$ = forward (k +1), we do not use k+1
- if EBITDA = current (t), we use k+1

- k = cash conversion ratio (% EBITDA → CF)
- r = required return (cost of capital)
- g = growth rate of EBITDA or CF

Determining the value of the company – market multiple

Using EBITDA

Consider:

- EBITDA (forward) = 10 M €
- k=50%
- r=15%
- g=5%

$$EV/EBITDA = \frac{0.5}{0.15 - 0.05} = \frac{0.5}{0.10} = 5.0x$$

$$EV = 10 \cdot 5 = 50 \text{ M€}$$

"5X" is the multiple of the EBITDA, which is the essence of the financial evaluation

Using CASH FLOWS

In case of Cash Flow

- r=15%
- g=5%

Cash flow forward $\frac{EV}{CF_{t+1}} = \frac{1}{r-g}$

Cash flow trailing $\frac{EV}{CF_t} = \frac{1+g}{r-g}$

Determining the value of a company

The value of the company can established also by indicating the Enterprise Value as an EBITDA Multiplier.
The EBITDA multiplier is base on market studies, innovation trajectories, projections of growth, market stability and contextual factors

Suppose:

- Expected EBITDA in Year 5 = €3M
- Market multiple = 5× EBITDA
- In case of no debt → EV = Equity value

$$Enterprise\ Value_{Year5} = 15M$$

$$Equity\ Value_{Year5} = 15M$$

Determining the value of a company

The Fund then indicates the required number of shares of the target companies to make the investment worthy

If the company will be worth €15M and the fund needs €11.14M:

$$Ownership = \frac{11.14}{15} = 74\%$$

The fund would require 74% of the company.

Now the key question: Is the Fund able to manage the majority of the firm? Are Fund managers entrepreneurs?

Is this acceptable for current shareholders?

Certain Fund decide ex-ante that the participation in the shares should be below 50% because the entrepreneur is the one that knows the market.

Determining the value of a company : Post money vs Pre money valuation

Definitions Pre money valuation: implied value of the venture before the investment.
For the investor it is the value of one share.

Post money valuation: totale value of the firm that is implied after the investment

$$Post\text{-}money = Pre\text{-}money + Investment$$

If the fund invests €3M for 74%:

$$Post\text{-}money = \frac{3}{0.74} = 4.05M$$ this means the 100% of the firm

$$Pre\text{-}money = 4.05 - 3 = 1.05M$$

For the investor, the investment makes the value of the firm increasing.
However, the entrepreneur may not agree on this evaluation..

The fund is implicitly valuing the company today at €1.05M.

Valuation drives ownership.

Determining the value of a company

Founders retain 26%.

If future value = €15M:

26% *15 M = 3.9 M --> For the founders the value of their shares would be 3.9 M

But what if they do not accept the fund?

Without external capital:

- Lower growth
- Lower EBITDA
- Lower exit value

Is the investment worthy for all the parties involved?
Diluting share with external investors requires the entrepreneur to determine whether the investor is ensuring growth

Decision rule:

$$(Ownership_{with\ fund} \times Value_{with\ fund}) > Value_{without\ fund}$$

Dilution is not destruction.
It depends on growth.

Investment decisions

- Is the company going to be acquired by the Fund?

It depends on:

- Is cash projection stable and feasible?
- IRR vs Promised returns to the investors. If IRR is higher → investment may be done
- Share of the company:
  - if it's necessary to get higher shares of a company to match target IRR, the Fund managers need to ask itself if they can run the company (fund managers are not entrepreneurs)
  - If less than 50% of shares are ok to make the investment worthy, fund managers and entrepreneurs need to agree on the business plans, the assumptions and how to make cash flows being respected over the years

Short term decisions vs long-term strategic decisions

Break-even told us:

- Minimum scale for survival

But investors care about:

- Growth trajectory
- Scalability
- Exit potential

Break-even = operational threshold
IRR = capital return threshold

Different metrics.
Different objectives → short term vs long term